Computer Architecture and Operating Systems
COSC 22073/BECS 22233

Lesson 02: Overview of the Computer System

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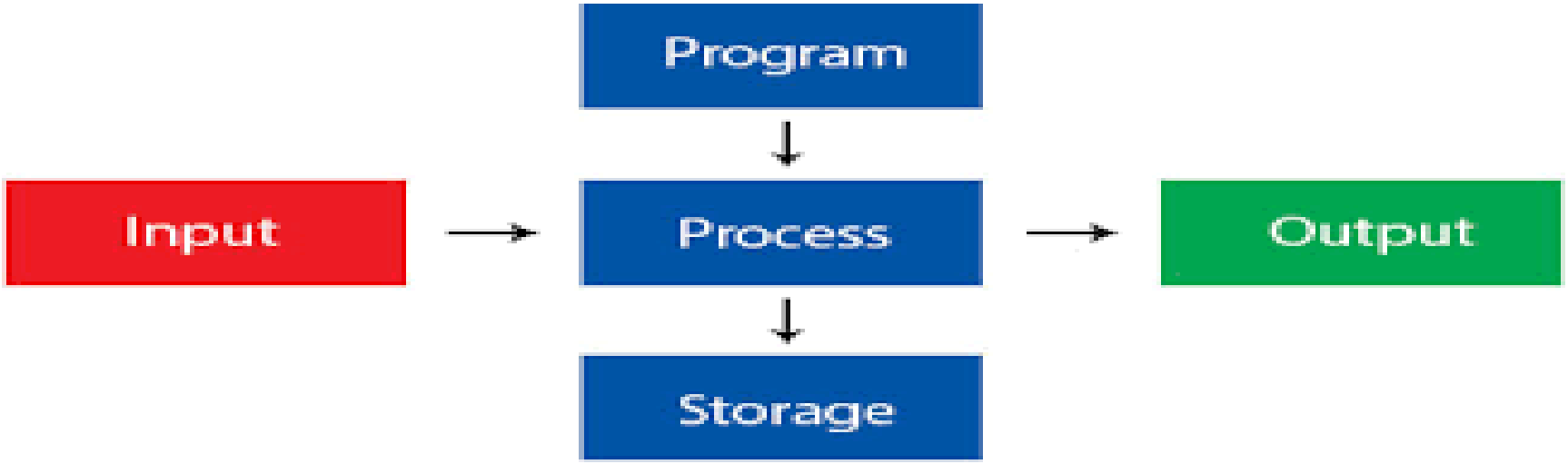
What is a Computer?

- A fast electronic calculating machine that accepts digitized input information, processes it according to a list of internally stored (memory) instructions (Program), and produces the resulting output information.



Functionalities of a computer

Any digital computer carries out five functions in gross terms:

- Takes data as input.
- Stores the data/instructions in its memory and use them when required.
- Processes the data and converts it into useful information.
- Generates the output
- Controls all the above four steps.

Classes of Computers (Based on Size)

- Supercomputers
- Mainframe Computers
- Minicomputers
- Desktop Computers
- Laptops (Notebook Computers)
- Tablets
- Ultrabooks
- Mini PCs

Classes of Computers (Based on Functionality)

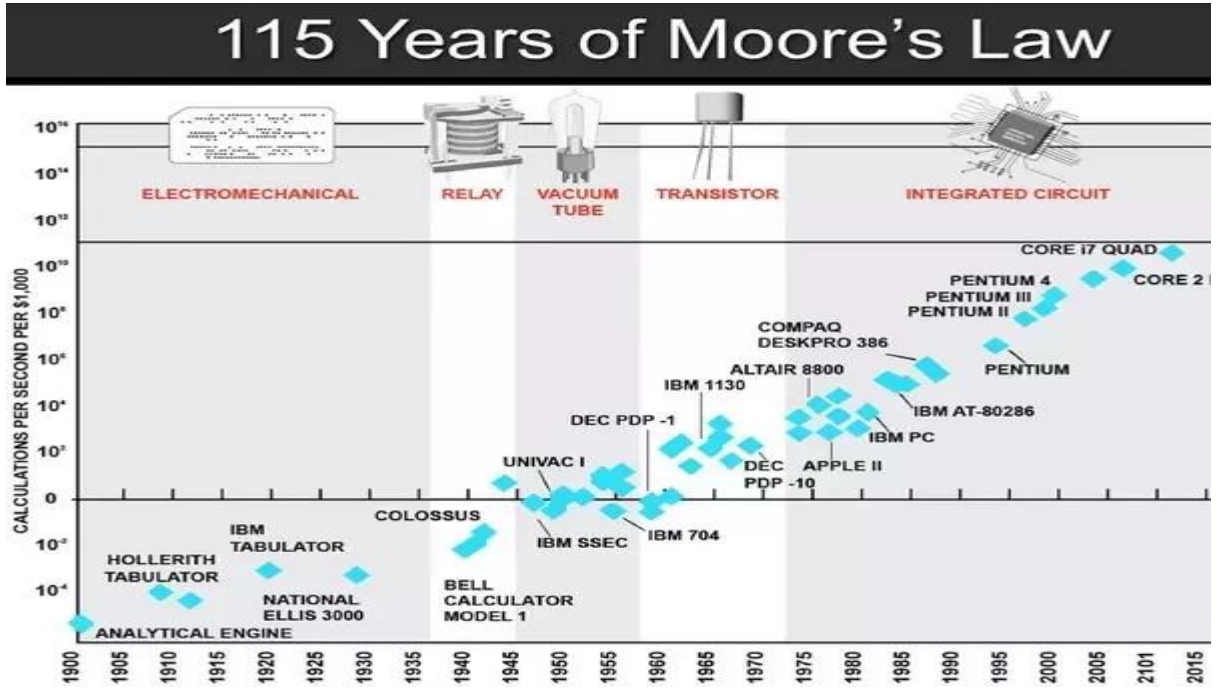
- General-Purpose Computers
- Special-Purpose Computers
- Server Computers
- Embedded Computers
- Cloud Servers and Data Centers
- Supercomputers
- Mainframe Computers
- Workstations
- Thin Clients
- Gaming Consoles
- Smartphones and Tablets
- Single-Board Computers (SBCs)
- Quantum Computers

Key features and concepts of quantum computers.

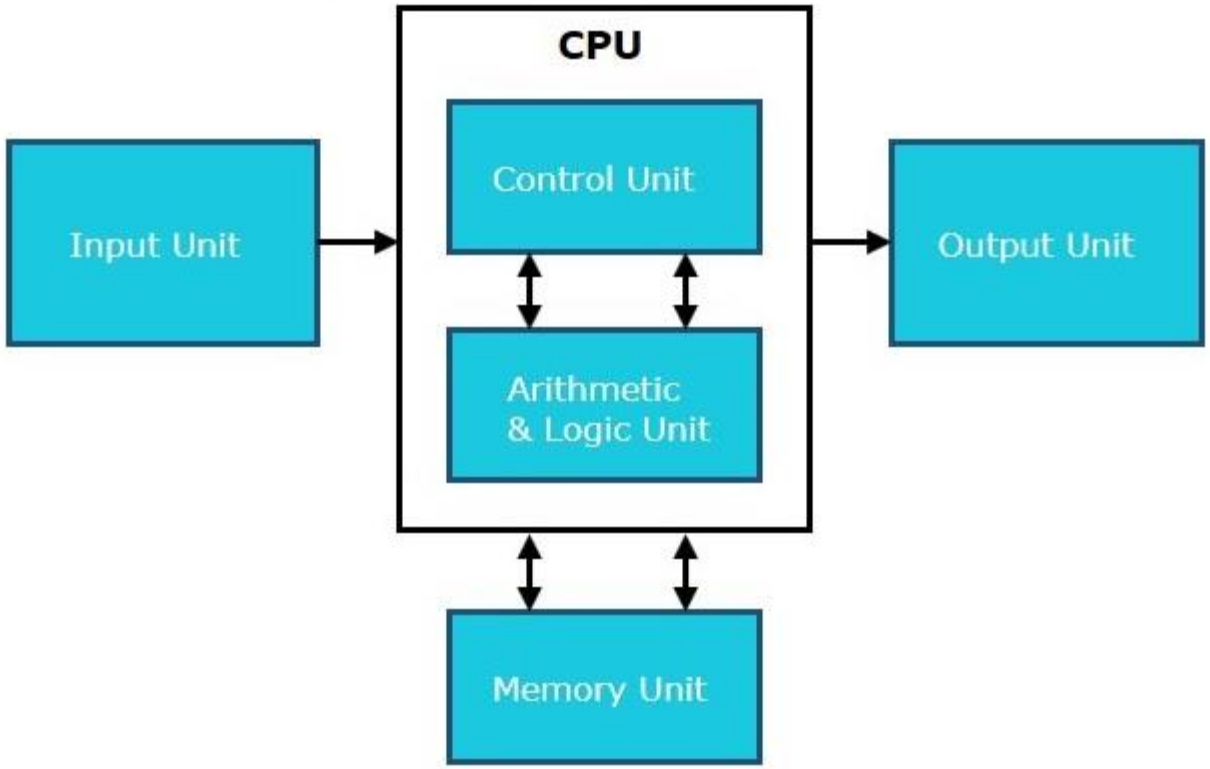
- Superposition — **A quantum bit (qubit) can exist in multiple states simultaneously (both 0 and 1 at the same time), unlike a classical bit, which is either 0 or 1.**
- Entanglement — **Qubits can become entangled, meaning the state of one qubit is instantly connected to the state of another, no matter how far apart they are.**
- Quantum Interference — **Quantum states can interfere with each other, amplifying the correct solutions while canceling out the wrong ones.**
- Quantum Parallelism — **By using superposition and entanglement, quantum computers can process many computations at once.**
- Quantum Tunneling — **A qubit can "tunnel" through energy barriers, meaning it can transition between states in ways that are impossible for classical systems.**
- Exponential Speedup — **This speedup enables quantum computers to solve problems that would take classical computers billions of years**

Moore’s Law

- In 1965, Intel’s Gordon Moore predicted that the number of transistors that can be integrated on single chip would double about every two years



Organization of a Computer:
Four classic components of a computer – input, output, memory and processor.

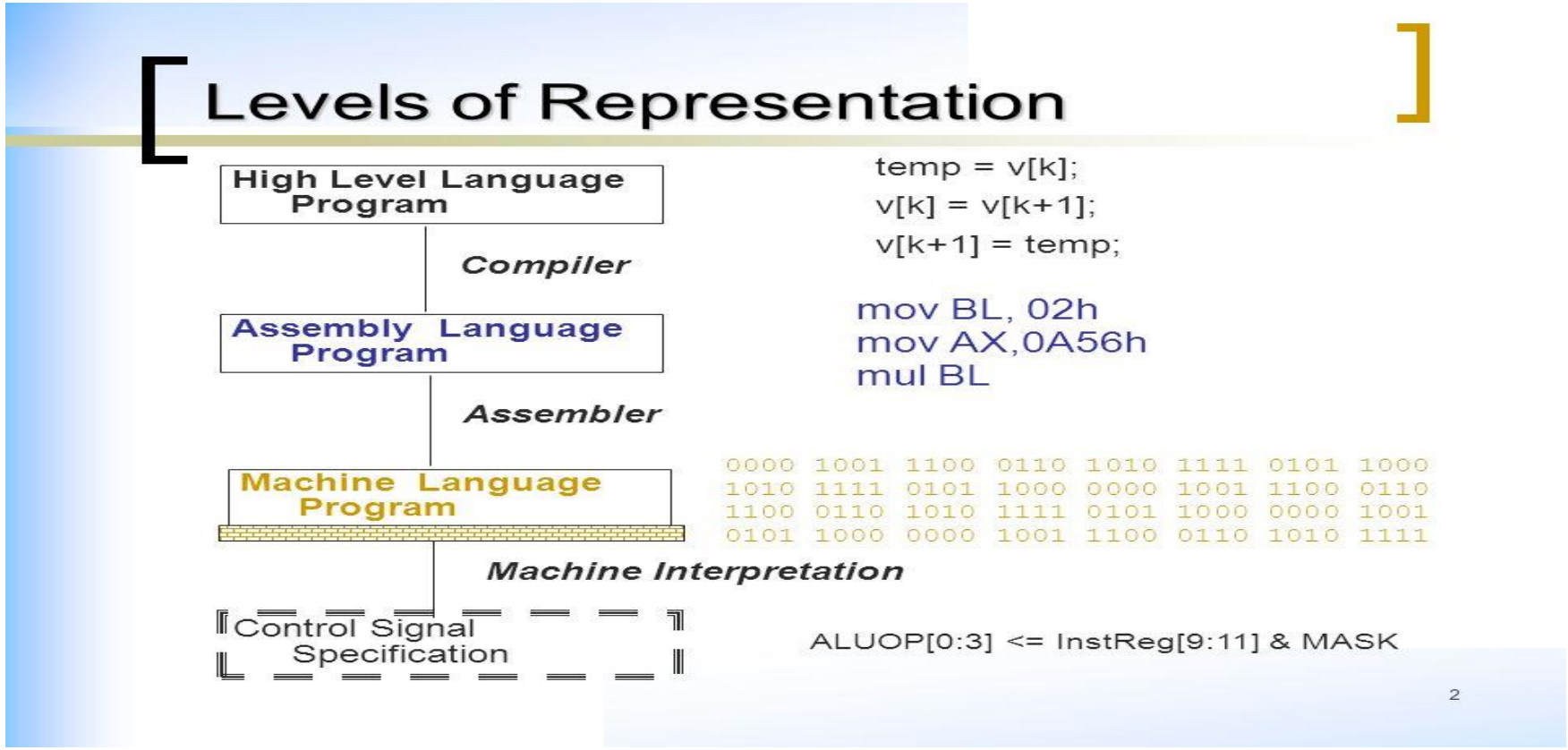


Levels of Representation

Each level represents a specific aspect of how a computer operates or how software interacts with hardware.

1. Physical Level **This is the hardware layer, which includes the physical components of a computer (e.g., transistors, resistors, wires, and circuits)**
2. Digital Logic Level **Physical components are abstracted into logic gates (AND, OR, NOT) and circuits**
3. Microarchitecture Level (Processor Level) **It describes how the processor is structured, including registers, ALUs (Arithmetic Logic Units), pipelines, and control units.**
4. Instruction Set Architecture (ISA) Level **Defines the set of instructions that a processor can execute. It serves as the interface between software and hardware.**
5. Assembly Language Level **A low-level programming language closely tied to the ISA.**
6. High-Level Programming Language Level **Compilers or interpreters convert high-level code into machine code or assembly code.**
7. Operating System Level **The operating system acts as a bridge between hardware and application software.**
8. Application Software Level **This is the highest level, where end-users interact with the system through applications (e.g., word processors, web browsers, games)**

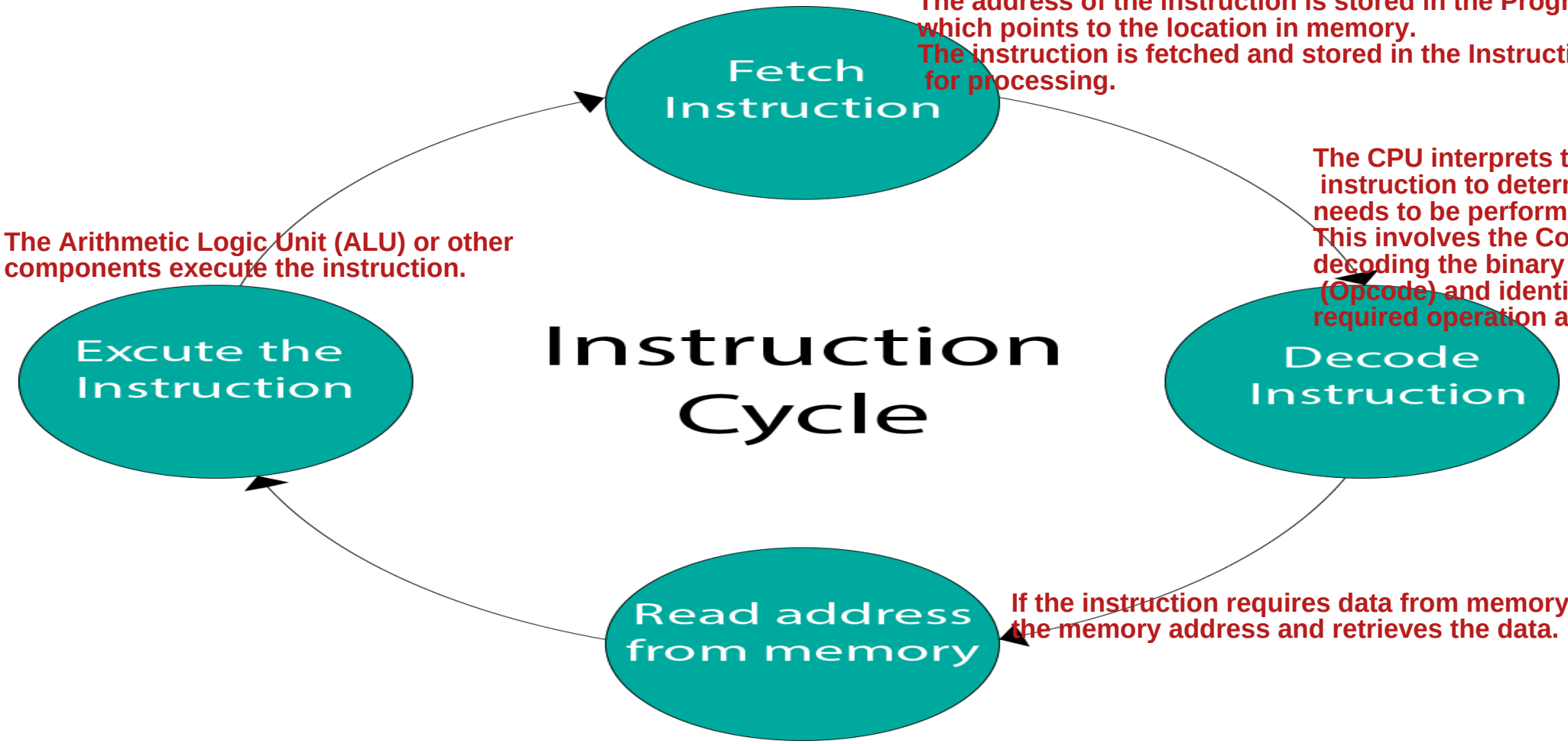
Levels of Representation



Instruction Execution Cycle :is a fundamental process in how a CPU executes instructions

The CPU retrieves the next instruction to execute from memory.
The address of the instruction is stored in the Program Counter (PC), which points to the location in memory.
The instruction is fetched and stored in the Instruction Register (IR) for processing.

The CPU interprets the fetched instruction to determine what operation needs to be performed.
This involves the Control Unit (CU) decoding the binary format of the instruction (Opcode) and identifying the required operation and operands



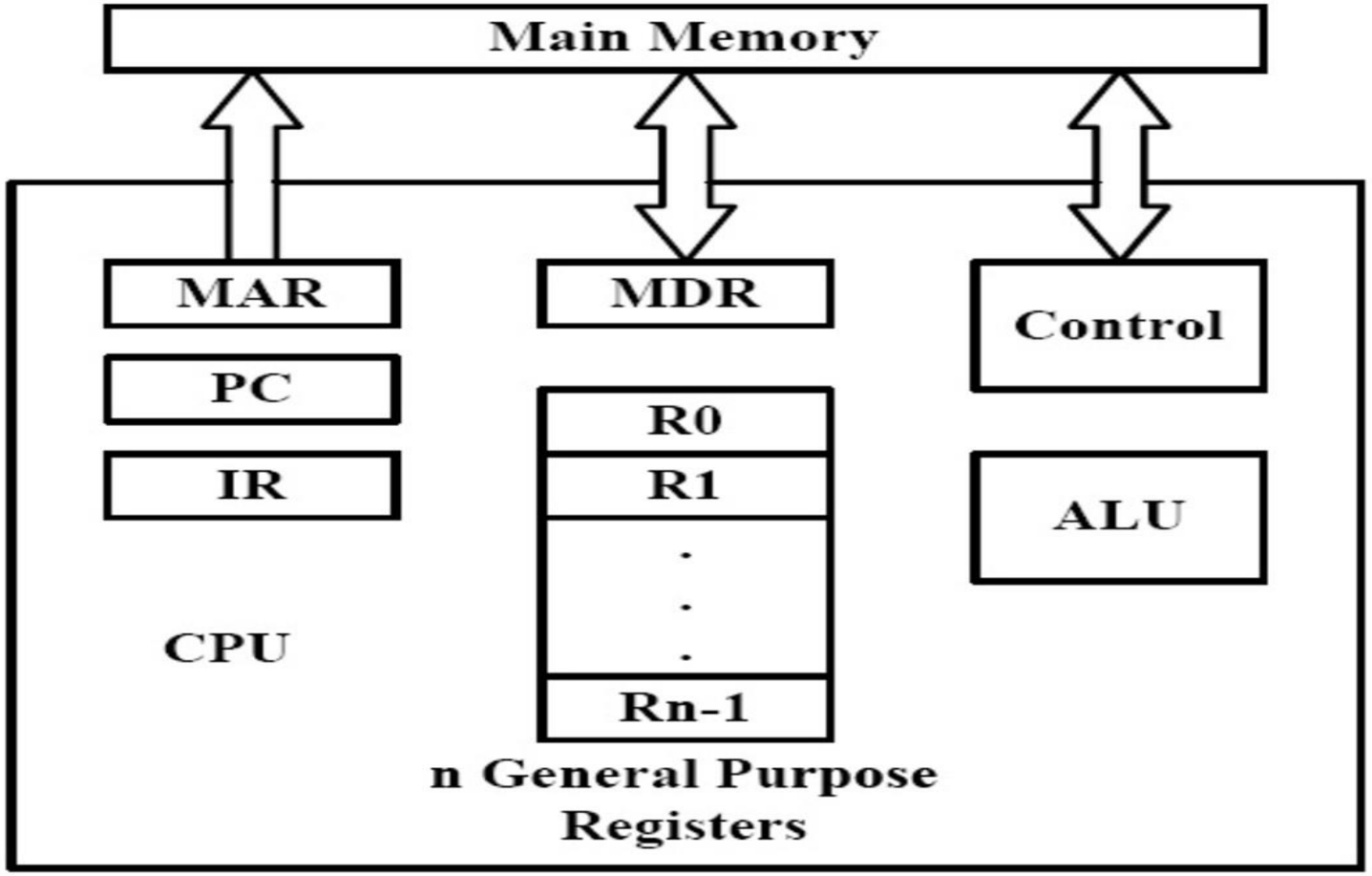
If the instruction requires data from memory, the CPU generates the memory address and retrieves the data.

The Arithmetic Logic Unit (ALU) or other components execute the instruction.

Instruction Execution Cycle

1. Obtain instruction from program storage.
2. Determine required action and instruction size.
3. Locate and obtain operand data.
4. Compute result value or status.
5. Deposit results in storage for later use.
6. Determine successor instruction.

Connection Between Processor and Memory



Program Counter (PC):

Holds the address of the next instruction to be fetched from memory.

Memory Address Register (MAR):

Temporarily stores the address in memory where data or an instruction is to be fetched or stored.

Memory Data Register (MDR):

Holds the data fetched from memory or data that is to be written into memory.

Instruction Register (IR):

Holds the fetched instruction currently being executed.

Exercises

1. List the steps needed to execute the machine instruction : Add R1,R2
2. List the steps needed to execute the machine instruction : Add R1,R2,R3

Thank You.